

For each cell i , compute Shannon entropy of its transition distribution:

$$\text{DEn}(i) = -\sum_j T(i,j) * \log(T(i,j))$$

DEn	Biological Meaning
High	Many possible destinations = progenitor/transitioning
Low	One clear destination = committed/terminal
Zero	Self-loop only = fully stable attractor

```
import numpy as np
from scipy.stats import entropy

def compute_den(transition_matrix):
    """
    Compute Differentiation Entropy per cell.
    High DEn = unstable/progenitor
    Low DEn = committed/terminal
    """
    den = np.apply_along_axis(entropy, 1, transition_matrix)
    return den
```

CellRank Implementation (existing tool)

CellRank is the main package implementing Markov-based cell fate analysis.

```
import cellrank as cr
import scvelo as scv

# Step 1: Compute RNA velocity
scv.tl.velocity(adata)
scv.tl.velocity_graph(adata)

# Step 2: Build transition matrix from velocity
vk = cr.kernels.VelocityKernel(adata)
vk.compute_transition_matrix()

# Optional: combine with expression similarity
ck = cr.kernels.ConnectivityKernel(adata)
ck.compute_transition_matrix()
combined = 0.8 * vk + 0.2 * ck
```

```

# Step 3: Compute macrostates (attractors)
estimator = cr.estimators.GPCCA(combined)
estimator.compute_macrostates(n_states=4) # naive, GC, plasmablast, plasma
estimator.plot_macrostates()

# Step 4: Identify terminal states (deep attractors)
estimator.compute_terminal_states()
estimator.plot_terminal_states()

# Step 5: Absorption probabilities (fate probability per cell)
estimator.compute_absorption_probabilities()
# adata.obsm['lineages_fwd'] = fate probability matrix

# Step 6: Differentiation Entropy (built into CellRank)
# High DEn = progenitor, Low DEn = committed
adata.obs['DEn'] = compute_den(
    vk.transition_matrix.toarray()
)

```

Integration with IsoDecipher APA Stability

The power comes from combining two INDEPENDENT methods:

```

def combined_stability_score(adata, gene_list):
    """
    Combine Markov-based DEn with APA-based stability.

    Agreement between methods = stronger biological evidence.

    Quadrants:
    - Low DEn + Low APA variance = TRUE terminal cell (deep attractor)
    - High DEn + High APA variance = TRUE transitioning cell
    - Low DEn + High APA variance = APA still resolving (interesting!)
    - High DEn + Low APA variance = committed APA, not committed fate (interesting)
    """
    # Normalize DEn to 0-1
    den = adata.obs['DEn'].values
    den_norm = (den - den.min()) / (den.max() - den.min())

    # Compute APA instability (variance across neighbors)
    pui_cols = [f'{g}_PUI' for g in gene_list
                 if f'{g}_PUI' in adata.obs.columns]

    if pui_cols:

```

```

from sklearn.neighbors import NearestNeighbors
pui_matrix = adata.obs[pui_cols].fillna(0.5).values

nn = NearestNeighbors(n_neighbors=15)
nn.fit(adata.obsm['X_umap'])
_, indices = nn.kneighbors(adata.obsm['X_umap'])

apa_var = np.array([
    np.nanmean(np.nanvar(pui_matrix[indices[i]], axis=0))
    for i in range(adata.n_obs)
])
apa_var_norm = (apa_var - apa_var.min()) / (apa_var.max() - apa_var.min())
else:
    apa_var_norm = np.zeros(adata.n_obs)

# Combined instability score
adata.obs['markov_instability'] = den_norm
adata.obs['apa_instability'] = apa_var_norm
adata.obs['combined_instability'] = (den_norm + apa_var_norm) / 2
adata.obs['combined_stability'] = 1 - adata.obs['combined_instability']

# Quadrant classification
threshold = 0.5
conditions = [
    (den_norm < threshold) & (apa_var_norm < threshold),
    (den_norm >= threshold) & (apa_var_norm >= threshold),
    (den_norm < threshold) & (apa_var_norm >= threshold),
    (den_norm >= threshold) & (apa_var_norm < threshold),
]
labels = [
    'Terminal (both agree)',
    'Transitioning (both agree)',
    'APA resolving, fate committed',
    'Fate plastic, APA committed'
]
adata.obs['stability_class'] = np.select(conditions, labels, 'Unknown')

return adata

```

Visualization: Stability Map

```

import matplotlib.pyplot as plt
import seaborn as sns

```

```

def plot_stability_quadrant(adata, color_by='cell_type'):
    """
    Scatter plot: DEn (x) vs APA instability (y)
    Color by cell type or stability class
    """
    fig, ax = plt.subplots(figsize=(8, 8))

    scatter = ax.scatter(
        adata.obs['markov_instability'],
        adata.obs['apa_instability'],
        c=adata.obs[color_by].astype('category').cat.codes,
        alpha=0.5, s=10, cmap='tab20'
    )

    # Quadrant lines
    ax.axvline(0.5, color='gray', linestyle='--', alpha=0.5)
    ax.axhline(0.5, color='gray', linestyle='--', alpha=0.5)

    # Quadrant labels
    ax.text(0.1, 0.1, 'Terminal\n(stable)', transform=ax.transAxes,
           fontsize=11, color='darkgreen', fontweight='bold')
    ax.text(0.6, 0.6, 'Transitioning\n(unstable)', transform=ax.transAxes,
           fontsize=11, color='darkred', fontweight='bold')
    ax.text(0.6, 0.1, 'Fate committed\nAPA resolving', transform=ax.transAxes,
           fontsize=9, color='darkorange')
    ax.text(0.1, 0.6, 'APA committed\nFate plastic', transform=ax.transAxes,
           fontsize=9, color='steelblue')

    ax.set_xlabel('Markov Instability (DEn)', fontsize=12)
    ax.set_ylabel('APA Instability (PUI variance)', fontsize=12)
    ax.set_title('Cell State Stability: Markov vs APA', fontsize=13)

    plt.tight_layout()
    return fig

```

Biological Interpretation of Quadrants

Quadrant	DEn	APA variance	Interpretation
Terminal	Low	Low	Deep attractor — fully committed
Transitioning	High	High	Saddle point — both fate and APA changing
APA resolving	Low	High	Fate decided, APA catching up (lag!)

Fate plastic	High	Low	APA committed first, fate still open
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The most interesting biology:

The “APA resolving” and “Fate plastic” quadrants reveal the **temporal ordering** of molecular events during commitment:

- If APA resolving is common → APA switching happens AFTER fate commitment
- If Fate plastic is common → APA switching is an EARLY event, predicts fate

This is a testable hypothesis unique to combining CellRank + IsoDecipher.

Connection to Bifurcation Framework

Markov Transition Matrix

↓

Identify bifurcation point (where DEn peaks along pseudotime)

↓

At bifurcation: does PUI bimodality coefficient also peak?

↓

YES → APA switching IS coupled to fate bifurcation

NO → APA switching is independent of fate decision

```
def correlate_den_bimodality(adata, gene, n_bins=25):
    """
    Test whether DEn peaks coincide with APA bimodality peaks.
    If correlated → APA switching coupled to fate bifurcation.
    """
    from scipy.stats import pearsonr, spearmanr

    pt = adata.obs['pseudotime'].values
    den = adata.obs['DEn'].values
    pui = adata.obs[f'{gene}_PUI'].values

    order = np.argsort(pt)
    step = len(pt) // n_bins

    bin_den = []
    bin_bc = []

    for i in range(0, len(pt) - step, step):
        idx = order[i:i+step]
```

```
window_pui = pui[idx]
window_den = den[idx]

from scipy.stats import skew, kurtosis
n = len(window_pui)
s = skew(window_pui)
k = kurtosis(window_pui)
bc = (s**2 + 1) / (k + 3*(n-1)**2/((n-2)*(n-3)) + 1e-10)

bin_bc.append(bc)
bin_den.append(np.mean(window_den))

rho, pval = spearmanr(bin_den, bin_bc)
print(f"DEn vs APA bimodality correlation: rho={rho:.3f}, p={pval:.4f}")

if rho > 0 and pval < 0.05:
    print("→ APA switching IS coupled to fate bifurcation")
else:
    print("→ APA switching is independent of fate decision")

return rho, pval
```

Available Tools Summary

Tool	What it does	Relevant for IsoDecipher
CellRank	Markov transition matrix, DEn, terminal states	Core integration
scVelo	RNA velocity, latent time	Pseudotime axis
Palantir	Markov fate probabilities, entropy	Alternative to CellRank
STEMNET	Fate probability estimation	Alternative
IsoDecipher	APA/PUI quantification	Your tool

The novelty: No existing tool combines Markov fate analysis with APA dynamics. This combination is unique to your framework.

Implementation Order (週一開始)

1. PUI notebook (Day 1)
2. APA trajectory + JSD (Day 2)
3. Bifurcation detection (Day 3)
4. CellRank integration + DEn (Day 4)
5. Combined stability quadrant plot (Day 5)
6. Waddington landscape visualization (Day 6)
7. bioRxiv preprint draft (Week 2)